WORKED SOLUTION

STEP II 2011, Question 12

Probability and Statistics • Geometric Series of Games

STEP II 2011, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q12

The key idea

Each game is only three points long, so we can list every way a game can go (there are just eight sequences) and sort them into "Xavier wins", "Younis wins" and "nobody wins, play another game". Because each new game starts afresh, the match is a geometric process: the probability Younis wins the match is his chance of winning a game divided by the chance that a game is decisive.

Solution

Write each game as a string of point-winners, so XYY means Xavier won the first point and Younis the second and third. The first point goes to X with probability p. After that, each point goes to the *same* player as the previous point with probability p and *switches* with probability $1-p$.

A game is decided as soon as the same player wins two points in a row, so the possible games are:

Sequence	Probability	Outcome
XX	$p \cdot p$	Xavier wins
YXX	$(1-p)(1-p)p$	Xavier wins
YY	$(1-p)p$	Younis wins
XYY	$p(1-p)p$	Younis wins
XYX	$p(1-p)(1-p)$	another game
YXY	$(1-p)(1-p)(1-p)$	another game

(Once a game reaches XX or YY the third point isn't needed, and it makes no difference whether we imagine it being played.) The undecided games are exactly those where the points alternate, i.e. both later points "switch".

Part (i)

Let A be the event that Younis wins a given game and C the event that the game is undecided. From the table,

$$\mathrm{P}(A) = (1-p)p + p^2(1-p) = p(1-p)(1+p) = p(1-p^2),$$
$$\mathrm{P}(C) = p(1-p)^2 + (1-p)^3 = (1-p)^2(p+(1-p)) = (1-p)^2.$$

Games are played independently with the same rules, so Younis wins the match if the first $n-1$ games are undecided and he wins game n, for some $n \geq 1$. Hence

$$w = \sum_{n=1}^{\infty} \mathrm{P}(C)^{\,n-1}\,\mathrm{P}(A) = \frac{\mathrm{P}(A)}{1-\mathrm{P}(C)},$$

where the geometric series converges because $\mathrm{P}(C) = (1-p)^2 < 1$ when $p \neq 0$. (Equivalently: if the first game is undecided we are back where we started, so $w = \mathrm{P}(A) + \mathrm{P}(C)\,w$.) Now $1-(1-p)^2 = 2p - p^2 = p(2-p)$, so

$$w = \frac{p(1-p^2)}{p(2-p)} = \frac{1-p^2}{2-p},$$

where cancelling p uses $p \neq 0$. This is the required result.

Comparing with $\frac{1}{2}$. We have

$$w - \tfrac{1}{2} = \frac{2(1-p^2)-(2-p)}{2(2-p)} = \frac{p-2p^2}{2(2-p)} = \frac{p(1-2p)}{2(2-p)}.$$

For $0 < p \leq 1$ the factors p and $2-p$ are positive, so $w - \frac{1}{2}$ has the sign of $1-2p$. Hence $w > \frac{1}{2}$ when $0 < p < \frac{1}{2}$ and $w < \frac{1}{2}$ when $p > \frac{1}{2}$ (and $w = \frac{1}{2}$ at $p = \frac{1}{2}$, as symmetry demands).

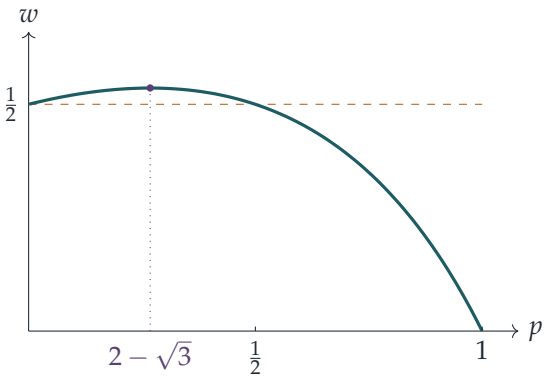
> **Ask yourself** Lowering p makes Younis more likely to win the first point. Must it always help him?

Is w decreasing in p? No. Differentiating with the quotient rule,

$$\frac{\mathrm{d}w}{\mathrm{d}p} = \frac{-2p(2-p)+(1-p^2)}{(2-p)^2} = \frac{p^2-4p+1}{(2-p)^2}.$$

The numerator has roots $p = 2 \pm \sqrt{3}$, and only $2-\sqrt{3} \approx 0.268$ lies in $[0,1]$. So $\mathrm{d}w/\mathrm{d}p > 0$ for $0 < p < 2-\sqrt{3}$ and $\mathrm{d}w/\mathrm{d}p < 0$ for $2-\sqrt{3} < p \leq 1$. On the interval $0 < p < 2-\sqrt{3}$, decreasing p actually *decreases* w. For example, $w(0.2) = \frac{8}{15} \approx 0.533$ but $w(0.1) = \frac{99}{190} \approx 0.521$.

The reason is that p plays two roles. Small p gives Younis the first point, but it also makes winning streaks unlikely, since a winner of a point keeps winning with probability only p. When p is tiny the points tend to alternate, and the advantage of the first point gets washed out. The largest value is $w(2-\sqrt{3}) = 4-2\sqrt{3} \approx 0.536$, and $w \to \frac{1}{2}$ as $p \to 0$.



Part (ii)

With $p = \frac{2}{3}$,

$$w = \frac{1 - \frac{4}{9}}{2 - \frac{2}{3}} = \frac{5/9}{4/3} = \frac{5}{12}.$$

Since $\mathrm{P}(C) = \frac{1}{9} < 1$, the match ends with probability 1, so Xavier wins it with probability $1 - \frac{5}{12} = \frac{7}{12}$. Xavier's expected gain is

$$\tfrac{7}{12} \times 1 - \tfrac{5}{12} \times k,$$

and the game is fair when this is zero, i.e. when

$$k = \frac{7}{5}.$$

So Xavier should pay Younis £1.40 when Younis wins.

Part (iii)

When $p = 0$, Younis wins the first point of every game for certain, and every later point is won for certain by the player who *lost* the previous point. So every game goes YXY with probability 1. Nobody ever wins two consecutive points, and the match never ends: neither player wins.

This is why part (i) excludes $p = 0$. There $\mathrm{P}(A) = 0$ and $1 - \mathrm{P}(C) = 0$, so the formula $w = \mathrm{P}(A)/(1 - \mathrm{P}(C))$ becomes $0/0$. The fact that $\frac{1-p^2}{2-p} \to \frac{1}{2}$ as $p \to 0$ describes nearby matches that do eventually end; it is not a probability for the case $p = 0$ itself.

Checks

- At $p = \frac{1}{2}$ every point is a fair coin toss, so w should be $\frac{1}{2}$. The formula gives $\frac{3/4}{3/2} = \frac{1}{2}$.
- At $p = 1$ Xavier wins the first two points for certain, so $w = 0$. The formula gives 0.
- The six game probabilities in the table add to 1.

- An exact enumeration of all eight point sequences (for p from 0.01 to 0.99) reproduced w, and a simulation of 200 000 matches at $p = \frac{2}{3}$ gave Younis winning 41.6% of the time, against $\frac{5}{12} \approx 41.7\%$.

Common slips

- Getting the transition rule the wrong way round. The probability p belongs to whoever won the *previous point*, not to Xavier.
- Answering $k = \frac{5}{7}$ by equating the wrong products. Fairness means $\mathrm{P}(\text{X wins}) \times 1 = \mathrm{P}(\text{Y wins}) \times k$, so the less likely winner gets the bigger payout.
- Plugging $p = 0$ into the formula in part (iii) and saying each player has probability $\frac{1}{2}$, when in fact the match never ends.

Beyond STEP

The points here form a *Markov chain*: each point depends only on the one before it. Splitting the match into identical, independent games is an example of a *renewal* argument, and the equation $w = \mathrm{P}(A) + \mathrm{P}(C)w$ is the simplest example of "first-step analysis", the standard method for absorption probabilities in Markov chains. Part (iii) shows what happens when absorption fails: the chain cycles for ever. Grimmett and Welsh, *Probability: An Introduction*, covers Markov chains and these methods at first-year level.

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